PyGameMaker — Tutorial 15: Fruit Fusion — Answer Key

> For teachers only — do not hand out to students. The numbering below matches this tutorial's Worksheet.

Worksheet Answer Key

Part A: 1-C, 2-D, 3-A, 4-B. **Part B:** 1. obj_player, Collision with obj_fruit_cherry. 2. obj_spawn_cherry, Alarm 0. 3. obj_fruit_cherry (and obj_fruit_strawberry, obj_fruit_orange), Outside Room. 4. obj_win_text, When drawing. 5. obj_win_text, Key press: space. **Part C:**

1. The collision's *If* branch only checks held_level == 3 (holding an orange); if held_level is something else, the *If* is false and the *Otherwise* branch runs instead — the fruit's own type never changes, only what you currently hold does, and a cherry can only ever match a basket already holding a cherry.
2. The watermelon fusion would never trigger: a correct match would still add 50 points and swap the sprite only if those actions were (also) left in the *If* branch, but *Go to room room_win* sitting in *Otherwise* would instead fire on every MISMATCH, sending the player to the win room without ever actually completing the fusion — the opposite of the intended behaviour.
3. One spawner per type means the collision event for that exact object (obj_fruit_cherry, say) already identifies the tier, with no extra conditional needed. A single random spawner would need the spawned fruit to carry its own "which tier am I" variable, and every collision check would need to read other's variable before comparing it to held_level — more moving parts for the same result.

Parts D and E: completion and reflection.